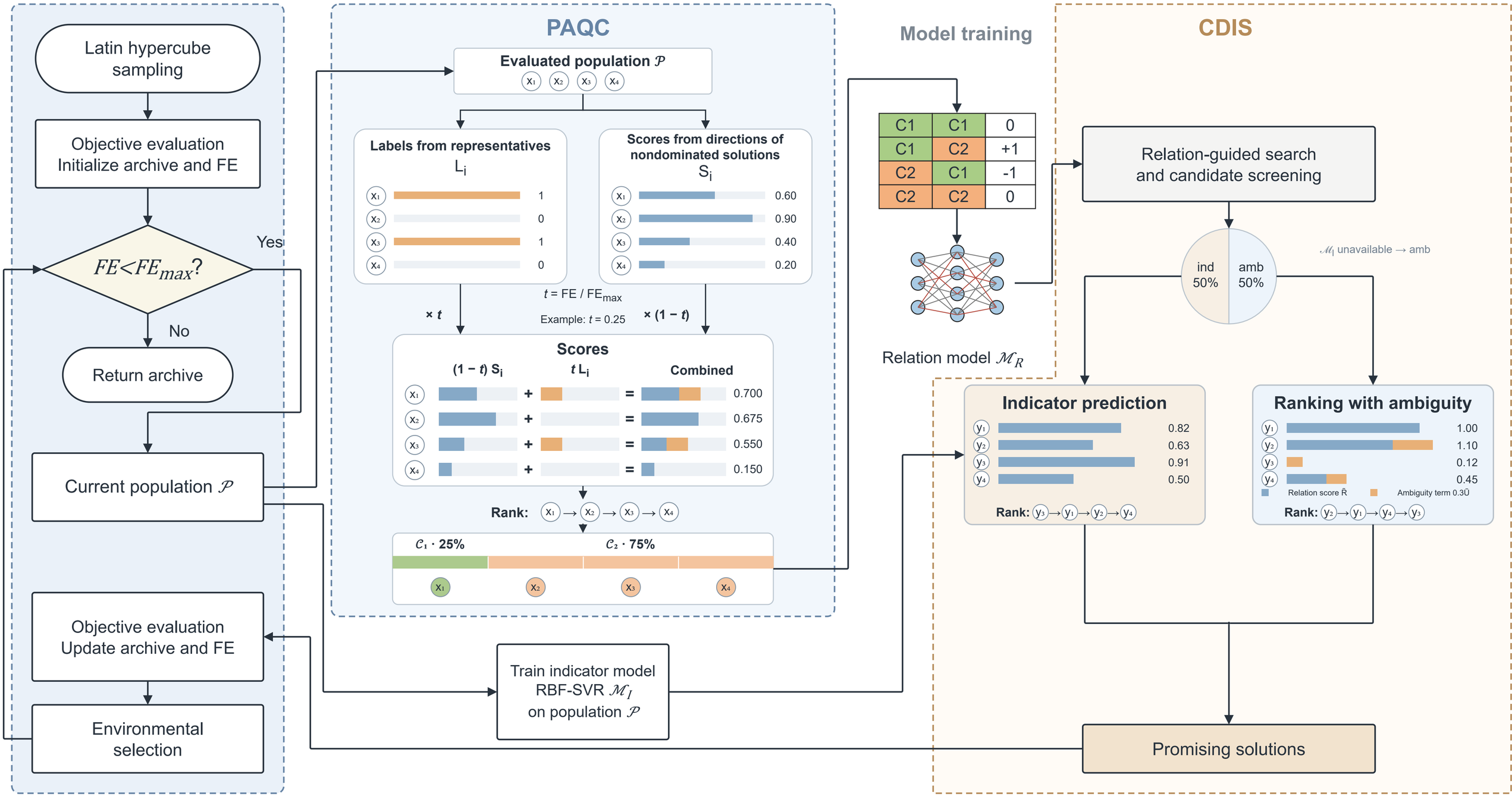

MOEA loop
PACDIS framework
Latin hypercube sampling
Objective evaluation
Initialize archive and FE
FE<FEmax?
Yes
No
Return archive
Current population P
Objective evaluation
Update archive and FE
Environmental selection
PAQC
Evaluated population P
x1 x2 x3 x4
Labels from representatives
Li
x1 1
x2 0
x3 1
x4 0
Scores from directions of nondominated solutions
Si
x1 0.60
x2 0.90
x3 0.40
x4 0.20
t = FE / FEmax
Example: t = 0.25
× t
× (1 − t)
Scores
(1 − t) Si
t Li
Combined
x1 + = 0.700
x2 + = 0.675
x3 + = 0.550
x4 + = 0.150
Rank: x1 → x2 → x3 → x4
C1 · 25%
C2 · 75%
x1 x2 x3 x4
Train indicator model
RBF-SVR MI
on population P
Model training
C1 C1 0
C1 C2 +1
C2 C1 -1
C2 C2 0
Relation model MR
CDIS
Relation-guided search
and candidate screening
ind 50%
amb 50%
MI unavailable → amb
Indicator prediction
y1 0.82
y2 0.63
y3 0.91
y4 0.50
Rank: y3 → y1 → y2 → y4
Ranking with ambiguity
y1 1.00
y2 1.10
y3 0.12
y4 0.45
Relation score R̄
Ambiguity term 0.3Ū
Rank: y2 → y1 → y4 → y3
Promising solutions